

claude-windows / 78dabe12-41fd-46c2-ae90-ab74bdaec3c6

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\78dabe12-41fd-46c2-ae90-ab74bdaec3c6\subagents\workflows\wf\_18dcd7ff-5ae\agent-a0f7e1c1ed53b6adf.jsonl`
- SHA-256: `a1a40ab437e8799d058dc34f1a5a56f6768930d0a5df33d95cebb00b6fb3cd2a`
- Source modified: `2026-06-22T10:26:45+00:00`
- Imported at: `2026-07-05T16:48:26+00:00`
- project: `wf\_18dcd7ff-5ae`
- session_id: `78dabe12-41fd-46c2-ae90-ab74bdaec3c6`

Transcript

1. user (2026-06-22T10:25:19.419Z)

CONTEXT ? a GCS corpus-migration work item just completed on branch work/golden-corpus-hygiene (PR #316, repo Khelsutra/badminton-highlight-indexer). Goal: run ALL training+inference from the cloud with ALL data in gs://khelsutra-rally-corpus so the local Windows box + external drives stop being blockers.

AUTHORITATIVE DOC TO REVIEW: docs/DATA_IN_GCS.md (the corpus source-of-truth map).
RELEVANT CODE: backend/worker/__main__.py (cmd_infer ~157, cmd_train ~332, --corpus-uri ~473), backend/storage/ref.py (URI schemes/resolve_input_uri), deploy/gcp/README.md, docs/SETUP_NEW_MACHINE.md, docs/archives/past_projects/DECOUPLED_COMPUTE/.

VERIFIED BUCKET STATE (already confirmed deterministically this session ? treat as ground truth, do NOT

re-run gcloud, it may not be authed for you):

- trajectories/ = 15 objects (<name>.csv)
- labels/ = 15 canonical objects only, named <YYYY-MM-DD>_<name>.rallies.csv (date = earliest mtime across C:/F: copies); the 7 old inconsistent (*_proxy / bare GX010128,mahadevpura-1) were MD5-verified byte-identical dupes and DELETED.
- manifests/golden.json = 15 entries, every trajectory+label gs:// ref RESOLVES (0 dangling).
- videos/ = mahadevpura_1.mp4 (523MB, demo of Drive->bucket) + 3 pre-existing (GX030094_proxy.mp4, mahadevpura-1.MP4, mahadevpura_smoke_180s.mp4). 14 of 15 golden videos NOT yet uploaded.
- worker train reads labels/(+videos/) and BUILDS the manifest on-box (does NOT read manifests/golden.json); manifests/golden.json is for EVAL. trajectories are a cache (training can regenerate via --trajectory-source detector).
- Open: Phase-2 videos (7 Drive proxies + 8 large F: full-MP4s, proxy-first), Phase-3 flip eval/litmus to GCS.
- Sibling threads (from memory): a golden-fixtures P3 thread needs these same golden traj+label CSVs; the served Gate-A litmus (#236, merged) is pinned to local paths; original-6 labels turned out recoverable from F:.

LENS = NEXT-SESSION FORESIGHT. Synthesize a PRIORITIZED "what to watch for / pick up next" list for the next session, spanning: PR #316 (open, review/merge), Phase-2 (rclone-on-cloud vs owner-box, proxy-first), Phase-3 (flip eval/litmus to GCS + the #236 re-measure-on-15 follow-up), the sibling golden-fixtures P3 thread that needs these same CSVs (now in GCS), and the shared-checkout/path-fragility hazards. Be concrete and ordered by what unblocks the most.

Return ONLY the structured verdict.

2. assistant (2026-06-22T10:25:22.500Z)

I'll review the authoritative doc and relevant code to ground my foresight synthesis in the actual current state.

3. user (2026-06-22T10:25:23.451Z)

```
1      # Data in GCS ? the corpus source-of-truth map
2
3      **Status:** LIVE / in-progress (migration not complete). **Owner-gated** decisions in
$7.
4      **Last inventoried:** 2026-06-22 against `gs://khelsutra-rally-corpus`.
5      **Goal (owner, 2026-06-22):** run **all** training + inference from the cloud with **all
data in GCS**,
6      so the local Windows box (and the intermittently-mounted external drive) is **no longer
a blocker**.
7
8      > **? Other sessions: this is where the data lives.** Pull from `gs://khelsutra-rally-
corpus`, not
9      > from anyone's `C:` / `F:` / Google Drive. If something you need isn't in the bucket
yet, it's in the
10     > **gap table ($5)** ? flag it / migrate it, don't hard-code a local path.
11
12     This doc is the **data map**. The *machine* runbooks are
[`SETUP_NEW_MACHINE.md`](SETUP_NEW_MACHINE.md)
13     (spin up a box) and [`deploy/gcp/README.md`](../deploy/gcp/README.md) (provision GPU +
train/infer).
14     The cloud engine itself was built by the (archived) DECOUPLED_COMPUTE project
15     (`docs/archives/past_projects/DECOUPLED_COMPUTE/`); this doc does **not** redesign it ?
it records
16     *where the bytes are* and *what's still missing*.
17
18     ---
19
20     ## 1. The bucket
21
22     | | |
23     |---|---|
24     | Bucket | **`gs://khelsutra-rally-corpus`** (single corpus bucket) |
25     | Project | `gen-lang-client-0306026826` (billing *Khelsutra Billing*) |
26     | Region | `asia-south1` (Mumbai), UBLA, co-located with the GPU VMs |
27     | Auth | SA `rally-worker@gen-lang-client-0306026826.iam.gserviceaccount.com`
(`roles/storage.objectAdmin`); key at `~/gcp/rally-worker-sa.json` (local only, **never
committed**). Interactive: `gcloud auth login`. |
28     | Size today | ~2.06 GB (mostly past dry-run reels/outputs ? **not** the full corpus) |
29
30     The engine is **backend-agnostic by URI** (`backend/storage/ref.py`): `gs://` (or
`gs://`),
31     `s3://`, `gdrive://`, or a bare local path. Pick the backend with **config + creds, no
code change**.
32
33     ---
34
35     ## 2. Canonical layout (the target contract)
36
37     The cloud worker's input/output contract (`backend/worker/__main__.py`; lineage:
38     `DECOUPLED_COMPUTE/04`):
39
40     | Prefix | Holds | Who reads/writes it |
41     |---|---|---|
42     | `videos/` | source videos / proxies (<clip>.mp4) | **input** to `worker infer` &
`worker train` |
43     | `labels/` | golden rally labels (<clip>.rallies.csv) | **input** ground truth to
```

```

`worker train` + all eval |
44 | `trajectories/` | cached WASB trajectory CSVs | **(PROPOSED ? see §7.2)** GPU-free
eval/litmus input |
45 | `weights/<version>/` | trained Gen-0 bundles (`weights.json` + a run `manifest.json`)
| **output** of `worker train` |
46 | `models/` | upstream pretrained weights (`wasb_badminton_best.pth.tar`) | input to the
WASB detector |
47 | `outputs/<video_id>/` | per-video inference outputs (intervals, etc.) | **output** of
`worker infer` |
48 | `highlights/`, `promo/` | rendered reels + promo artifacts | output / demo |
49
50 > **Note on the manifest.** `worker train` does **not** read a pre-staged corpus
`manifest.json` ? it
51 > *builds* one on-box at train time from `labels/` (+`videos/`), extracting trajectories
via the native
52 > detector (`--trajectory-source detector`). A `manifest.json` only appears as an
**output** inside each
53 > `weights/<version>/` bundle. The **eval/litmus** path is what wants a stable, path-
resolvable manifest
54 > (§7.3).
55
56 ---
57
58 ## 3. What's actually in the bucket today (2026-06-22)
59
60 > **Phase-1 landed 2026-06-22:** added `trajectories/` (15), 15 canonical
`labels/<date>_<name>.rallies.csv`,
61 > and `manifests/golden.json`. The table below is the **pre-Phase-1** snapshot + the
*remaining* gaps
62 > (videos, the 7 stale label objects to delete) ? see §7/§8.
63
64 | Prefix | Present | Drift / notes |
65 |---|---|---|
66 | `labels/` | 7 files: `GX010128`, `mahadevpura-1`, + `Badminton BXH 2_proxy`, `Boxhill
Doubles_proxy`, `GX020094_proxy`, `GX030094_proxy`, `mahadevpura-2_proxy` | ? **inconsistent
naming** (`_proxy` on some, not others); ? **missing** the original-6 labels + `GX010137`,
`GX010141` |
67 | `videos/` | 3: `GX030094_proxy.mp4`, `mahadevpura-1.MP4`, `mahadevpura_smoke_180s.mp4`
| ? corpus videos/proxies almost entirely absent |
68 | `trajectories/` | ? | ? **does not exist** (every trajectory is local-only) |
69 | `models/` | `wasb_badminton_best.pth.tar` | ? (? WASB-C3 weight-provenance still an
open legal item before any external share) |
70 | `weights/` | `0.1.0-l4/` | ? first real L4 train bundle |
71 | `outputs/` | one `/`, `parity/` | ? dry-run outputs |
72 | `highlights/`, `promo/` | reels + promo | ? |
73
74 **Drift summary:** the bucket has extra prefixes (`highlights/`, `promo/`, `models/`
alongside
75 `weights/`) and **inconsistent `labels/` naming**, and is **missing the bulk of the
corpus inputs**
76 (`videos/`, all `trajectories/`, most `labels/`). It reflects past *dry runs*, not a
complete corpus.
77
78 ---
79
80 ## 4. How the cloud consumes it
81
82 Provisioning + the verified commands live in
[`deploy/gcp/README.md`](../deploy/gcp/README.md) §3?§5.
83 The data-facing entry points (`backend/worker/__main__.py`):
84
85 ```bash
86 # config (no secrets in config; auth via GOOGLE_APPLICATION_CREDENTIALS):
87 # { "storage": { "backend": "local", "gcs": { "project": "gen-lang-client-0306026826"
} } }

```

```

88
89 # INFERENCE on one cloud video -> outputs/<video_id>/ in the bucket
90 python -m backend.worker infer \
91     --video-uri gcs://khelsutra-rally-corpus/videos/<clip>.mp4 \
92     --out-uri gcs://khelsutra-rally-corpus --segmenter wasb_hybrid --config config.json
93
94 # TRAIN from the bucket (reads labels/ [+ videos/], extracts trajectories, builds
manifest on-box,
95 # runs training.gen0.harness, pushes weights/<version>/):
96 RALLY_DETECTOR_IMPL=native python -m backend.worker train \
97     --corpus-uri gcs://khelsutra-rally-corpus --out-uri gcs://khelsutra-rally-corpus \
98     --version <v> --trajectory-source detector --segmenter wasb_hybrid --config
config.json
99     ```
100
101 A bare/relative path resolves under a configured `input_root`/`input_backend`
102 (`backend/storage/ref.py::resolve_input_uri`), so workflows can be written bucket-
relative.
103
104 ---
105
106 ## 5. The gap ? what's still local-only (why the box is still a blocker)
107
108 All of this is tiny except the originals ? the metadata that blocks cloud eval is <
25 MB total:
109
110 | Data | Where it is now | Size | In bucket? | Needed for |
111 |---|---|---|---|---|
112 | 20 WASB trajectory CSVs | `C:\?\Annotation Setup\Collect\Trajectories\` | 17.5 MB
| ? none | eval/litmus (cached), train (`--trajectory-source cached`) |
113 | golden manifest | `output/human_lovo_manifest.json` (C:-absolute paths) | 4.5 KB | ? |
eval corpus definition |
114 | original-6 labels | `F:[Khelsutra]?\Training\Golden Labelled\*.rallies.csv` | ~12 KB
| ? | train + eval |
115 | 7 new-clip labels | repo `output/*.rallies.csv` (gitignored) F: archive (backed
up 2026-06-22) | ~12 KB | ? partial/inconsistent | train + eval |
116 | source videos / proxies | F: archive + Google Drive golden folder | 100s of GB | ?
3 only | infer + train (real WASB extraction) |
117
118 > The corpus metadata (trajectories + labels + manifest, ~25 MB) is what actually
blocks GPU-free
119 > cloud eval and the served Gate-A litmus ? and it's trivially small to migrate. The
videos are the
120 > only large migration, and they already exist on Google Drive (the golden-corpus
folder), so they
121 > can go Drive ? bucket from a cloud box, keeping the local machine out of the path
entirely.
122
123 ---
124
125 ## 6. Migration plan (close the gap)
126
127 Phase 1 ? corpus metadata (?25 MB, do first, removes the eval blocker). Stage all
trajectories,
128 all labels, and a bucket-relative eval manifest into `gs://khelsutra-rally-corpus`.
Pending the §7
129 naming decision. Sketch:
130
131 ```bash
132 # trajectories (decision-free; absent today)
133 gcloud storage cp "C:\?\Collect\Trajectories\*.csv" gs://khelsutra-rally-
corpus/trajectories/
134 # labels (after the §7.1 naming decision) -> labels/<canonical-name>.rallies.csv
135 # eval manifest with gs:// (or bucket-relative) paths -> a stable manifest object (§7.3)
136 ```

```

```

137
138     **Phase 2 ? videos ? `videos/<name>.mp4`** (bucket name = the eval `name`, consistent
with
139     `trajectories/` + `labels/`). Sources (resolved 2026-06-22 ? see
`scratch/copy_videos_to_gcs.ps1`):
140
141     | Source | Clips | Size |
142     |---|---|---|
143     | Drive golden folder `~/Golden Label Videos Request - June 15/[Done] Video N/`
(**proxies**) | `mahadevpura_1/2`, `GX010128`, `GX020094`, `GX030094`, `Badminton_BXH_2`,
`Boxhill_Doubles` (7) | ~13.5 GB |
144     | F: archive (**full MP4s** ? proxy-first before upload) | `GX010137`, `GX010141`, +
original-6 (`adarsh_avi_singles`, `kushagra_singles`, `largetest_doubles`, `gbaaddy`,
`testlarge_short`, `mahadevpura_singles`) (8) | ~50 GB |
145
146     **Two ways to run it:**
147     - **Decoupled (preferred ? no local box):** `rclone` on a cloud VM with a Google-Drive
remote ?
148     `rclone copy gdrive:"<?/[Done] Video N>/<proxy>.mp4" :gcs:khelsutra-rally-
corpus/videos/<name>.mp4`
149     (one-time owner step: `rclone config` ? `gdrive` remote OAuth). Or `gdown <file-id>`
on the box for
150     the shared folder. This is the path that actually removes the local box.
151     - **Interim (today, owner box ? streams Drive/F: ? local ? GCS):** `pwsh
scratch/copy_videos_to_gcs.ps1
152     -Execute` (dry-run without `-Execute`; `-Only <name>` for one; skip-existing). Uses
local bandwidth.
153
154     **Priority:** the 7 Drive **proxies** first (already 1080p, sufficient for detection).
The 8 large F:
155     full MP4s should be **proxied to 1080p first** (cheaper storage + faster cloud decode)
before upload.
156     ? Two pre-existing non-canonical video objects (`videos/GX030094_proxy.mp4`,
`videos/mahadevpura-1.MP4`)
157     are superseded once `videos/GX030094.mp4` / `videos/mahadevpura_1.mp4` land ? verify-
dupe-then-delete
158     (same as the label cleanup). Track per-clip in $8.
159
160     **Phase 3 ? flip eval/training to read GCS by default.** Point the eval manifest +
litmus at the
161     bucket (env override or `input_root`) so a fresh box/CI needs no local corpus. Re-verify
the suite +
162     the served Gate-A litmus read from `gcs://` (and re-measure if the corpus set changes ?
see PR #236).
163
164     ---
165
166     ## 7. Decisions
167
168     1. **`labels/` naming ? LOCKED (owner, 2026-06-22):** `labels/<YYYY-MM-
DD>_<name>.rallies.csv`, where
169     `<name>` is the eval/manifest clip name and `<YYYY-MM-DD>` is the label's **authoring
date** =
170     the *earliest* mtime across all known copies (C: working copy + F: archive ? robust
to copy-resets;
171     a restored C: copy shows today, the F: archive keeps the real date). Fixes the
`_proxy` inconsistency
172     and disambiguates recycled GoPro names by date. *(Future, when videos are in the
bucket: optionally
173     add a `video_id`/MD5 sidecar for the strongest keying ? deferred, needs hashing each
video.)*
174     2. **Cache `trajectories/` ? YES, done 2026-06-22.** `trajectories/<name>.csv` cached so
cloud/CI eval +
175     the served Gate-A litmus run **GPU-free** (training can still regenerate via
`--trajectory-source detector`).

```

```

176 3. **Eval manifest in GCS ? done 2026-06-22.** `gs://khelsutra-rally-
corpus/manifests/golden.json`
177 (15 clips, `gs://` trajectory + label paths). The **local** litmus keeps its local-
path manifest;
178 flipping the litmus to read the GCS manifest is Phase 3.
179 4. **Originals migration scope ? OPEN** ? proxies-only vs full 4K, and how much of the
15-clip golden
180 set vs the wider archive. Source = Google Drive ? bucket from a cloud box (not the
local box).
181
182 > Phase-1 was run via a dry-run planner (resolve every clip's trajectory + label, stamp
the label's
183 > authoring date, build the `gs://` manifest) ? `gcloud storage cp` (planner kept in
gitignored
184 > `scratch/`; promote it to `scripts/` if corpus-sync recurs).
185 > ? **Cleanup DONE 2026-06-22:** the 7 pre-existing inconsistent label objects
(`*_proxy.rallies.csv`,
186 > bare `GX010128`/`mahadevpura-1`) were confirmed **byte-identical (MD5)** to the new
canonical names
187 > (content also on C:/F:) and **deleted** ? `labels/` now holds exactly the 15 canonical
objects.
188
189 > ? **gcloud gotcha (Phase-2):** `gcloud storage cp` treats `[...]` as a glob, so source
paths under
190 > `[Done] Video N/` (Drive) or `[Khelsutra] ?` (F:) fail with "matched no objects." The
owner-box
191 > script (`scratch/copy_videos_to_gcs.ps1`) stages such paths through a bracket-free
temp file; the
192 > `rclone` cloud path has no such issue.
193
194 ---
195
196 ## 8. Progress tracker
197
198 | Item | Status |
199 |---|---|
200 | Bucket + SA + provisioning (deploy/gcp) | ? built |
201 | Cloud worker `infer`/`train` by URI | ? built (`backend/worker`) |
202 | This data-map doc | ? 2026-06-22 |
203 | Phase 1: trajectories ? `trajectories/` (15) | ? 2026-06-22 |
204 | Phase 1: labels reconciled ? `labels/<date>_<name>` (15) | ? 2026-06-22 |
205 | Phase 1: eval manifest ? `manifests/golden.json` | ? 2026-06-22 |
206 | Phase 1: remove 7 stale `*_proxy`/bare label objects | ? 2026-06-22 (MD5-verified
dups, deleted) |
207 | Phase 2: corpus videos/proxies ? `videos/<name>.mp4` | ? mapping + script done;
**1/15** demo'd (`mahadevpura_1`); 14 remain ? run `rclone` on a cloud box (preferred) or the
owner-box script |
208 | Phase 2: proxy the 8 large F: full-MP4 clips before upload | ? |
209 | Phase 3: eval/litmus default to the GCS manifest | ? |
210
211 **Definition of done:** a fresh GPU box (or CI) can run inference, the eval suite, and
Gen-0 training
212 reading **only** from `gs://khelsutra-rally-corpus` ? no `C:` / `F:` / local-Drive path
required.
213
214 ---
215
216 ## Related
217 - [`SETUP_NEW_MACHINE.md`](SETUP_NEW_MACHINE.md) ? spin up a box (storage backends in
$7).
218 - [`deploy/gcp/README.md`](../deploy/gcp/README.md) ? GCP GPU provisioning + train/infer
($5).
219 - [`VIDEO_LOCALITY_MODEL.md`](VIDEO_LOCALITY_MODEL.md) .
[`PER_VIDEO_WORKSPACE.md`](PER_VIDEO_WORKSPACE.md) ? locality tiers + `outputs/<video_id>/`.
220 - `docs/archives/past_projects/DECOUPLED_COMPUTE/` ? the cloud-engine design lineage.

```

4. user (2026-06-22T10:25:23.681Z)

```

1      """Storage value objects + URI parsing for the pluggable storage layer.
2
3      A ``StorageRef`` is a backend-agnostic pointer to one object. URIs:
4
5          <bare path>                -> local    (today's behaviour; e.g. C:/v/x.mp4,
/data/x.mp4)
6          local://<path>             -> local    (local:///abs or local://rel/p)
7          s3://<bucket>/<key>        -> s3       (AWS + R2 + D0 Spaces + MinIO via
endpoint_url)
8          gcs://<bucket>/<key>      -> gcs       (gs:// accepted, normalized to gcs://)
9          gdrive://<path>           -> gdrive    (path under your locally-MOUNTED Google
Drive "My Drive")
10
11      No cloud SDKs are imported here ? this module is pure stdlib so ``import
backend.storage``
12      stays cheap.
13      """
14      from __future__ import annotations
15
16      import re
17      from dataclasses import dataclass
18      from datetime import datetime
19      from typing import Optional
20
21      _SCHEME_RE = re.compile(r"^[A-Za-z][A-Za-z0-9+.\-]*://")
22      _KNOWN = ("local", "s3", "gcs", "gdrive")
23
24
25      @dataclass(frozen=True)
26      class StorageStat:
27          """Lightweight object metadata (uniform across backends)."""
28          size: int
29          modified: Optional[datetime] = None
30          etag: Optional[str] = None
31          content_type: Optional[str] = None
32
33
34      @dataclass(frozen=True)
35      class StorageRef:
36          """A backend + a location. ``bucket`` is the S3/GCS bucket; for ``local`` and
``gdrive`` it is
37          empty and ``key`` carries the path (a filesystem path, or a path under the mounted
Drive)."""
38          backend: str
39          bucket: str
40          key: str
41          uri: str
42
43          @property
44          def name(self) -> str:
45              return self.key.rstrip("/").rsplit("/", 1)[-1]
46
47
48      def parse_uri(uri: str) -> StorageRef:
49          """Parse a storage URI (or a bare filesystem path) into a ``StorageRef``.
50
51          A string with no ``scheme://`` prefix (including Windows ``C:\\...`` paths, which
52          have no ``//``) is treated as a local filesystem path ? preserving today's
behaviour.
53          """
54          s = str(uri)

```

```

55     m = _SCHEME_RE.match(s)
56     if not m:
57         return StorageRef("local", "", s, f"local://{s}")
58     scheme = m.group(1).lower()
59     rest = s[m.end():]
60     if scheme == "gs":
61         scheme = "gcs"
62     if scheme in ("local", "gdrive"):
63         # Path-keyed, no bucket: local FS, or a path under the mounted Google Drive "My
Drive".
64         return StorageRef(scheme, "", rest, s)
65     if scheme in _KNOWN:
66         bucket, _, key = rest.partition("/")
67         return StorageRef(scheme, bucket, key, s)
68     raise ValueError(f"unsupported storage URI scheme: {scheme}://{ (in {s!r})}")
69
70
71     def resolve_input_uri(raw: str, *, input_backend: str = "local", input_root: str = "")
-> str:
72         """Resolve a user-supplied input reference into a storage URI
(COMPUTE_DECOUPLED_SERVING M2).
73
74         Precedence (first match wins):
75         1. An explicit ``scheme://`` on ``raw`` (``gs://``, ``s3://``, ``local://``,
``gdrive://``) ? as-is.
76         2. An ABSOLUTE local path (``/data/x.mp4``, ``C:\\?``, ``C:/?``) ? as-is (always
local).
77         3. A bare/relative name + a non-empty ``input_root`` ? joined under it
(``"gs://bucket/videos"`` + ``"x.mp4"`` ? ``"gs://bucket/videos/x.mp4"``).
78         4. A bare/relative name + ``input_backend != "local"`` ? prefixed (``"gcs"`` +
``"bucket/x"`` ?
80         ``"gcs://bucket/x"``).
81         5. Otherwise ? ``raw`` unchanged (a relative/bare LOCAL path ? today's behaviour).
82
83         So the default (``input_backend="local"`` , ``input_root=""``) is an identity
passthrough for every
84         path used today, preserving byte-for-byte behaviour.
85         """
86         s = str(raw)
87         if _SCHEME_RE.match(s):          # 1. explicit scheme wins
88             return s
89         # 2. an anchored LOCAL path is always local, never joined. Detect this OS-
INDEPENDENTLY ?
90         #   os.path.isabs is platform- AND Python-version-dependent (on Windows + py3.13 a
POSIX-style
91         #   "/data/x" is NOT absolute). Anchored == a leading "/" or "\\ ", or a Windows
drive prefix "X:".
92         if s.startswith((" /", "\\ ")) or (len(s) >= 2 and s[1] == ":" and s[0].isalpha()):
93             return s
94         if input_root:                    # 3. join a relative name under the configured
root
95             return input_root.rstrip("/") + "/" + s.lstrip("/")
96         if input_backend and input_backend != "local":    # 4. prefix a bucket/key with the
declared backend
97             return f"{input_backend}://{s}"
98         return s                          # 5. bare/relative local path ? unchanged
99

```

5. assistant (2026-06-22T10:25:26.826Z)

Now let me quickly verify the key code anchors mentioned in the context to ground my findings.

6. user (2026-06-22T10:25:28.258Z)

```

450     def build_parser() -> argparse.ArgumentParser:

```



```

451     p = argparse.ArgumentParser(prog="python -m backend.worker",
452                                description="Storage-aware worker: pull ? run pipeline ?
push.")
453     p.add_argument("-v", "--verbose", action="store_true",
454                   help="DEBUG logging (the native detector command, frame counts, per-
stage detail)")
455     sub = p.add_subparsers(dest="cmd", required=True)
456
457     pi = sub.add_parser("infer", help="run inference on one video URI and push outputs")
458     pi.add_argument("--video-uri", required=True, help="local path / local:// / s3:// /
gcs://")
459     pi.add_argument("--out-uri", required=True, help="output prefix
(outputs/<video_id>/? is appended)")
460     pi.add_argument("--segmenter", default=None, help="default:
indexing.default_segmenter")
461     pi.add_argument("--config", default=None, help="config.json path (default:
./config.json)")
462     pi.add_argument("--scratch", default=None, help="local scratch dir (default: a temp
dir)")
463     pi.add_argument("--max-frames", type=int, default=None)
464     pi.add_argument("--done-uri", default=None,
465                   help="M6: storage URI for a completion MARKER (video_id + rc)
written on a "
466                   "successful run, so a remote (Cloud Run) dispatcher's bucket-
poll terminates. "
467                   "Default-OFF: omit it for the normal local run.")
468     pi.add_argument("--set", action="append", default=[], metavar="k=v",
469                   help="config override, e.g. --set indexing.skip_ai_handoff=true")
470     pi.set_defaults(func=cmd_infer)
471
472     pt = sub.add_parser("train", help="Gen-0 train-from-bucket (labels [+videos] ->
trajectories -> harness)")
473     pt.add_argument("--corpus-uri", required=True, help="prefix containing labels/ (+
videos/ for "
474                   "--trajectory-source detector), e.g. s3://bucket or gcs://bucket")
475     pt.add_argument("--out-uri", required=True, help="output prefix (weights/<version>/
is appended)")
476     pt.add_argument("--version", required=True, help="semver for the artifact bundle")
477     pt.add_argument("--trajectory-source", choices=["synth", "detector"],
478                   default="synth",
479                   help="synth = CPU-mock label-consistent trajectories (default, no
GPU); "
480                   "detector = REAL shuttle trajectories via the resolved
DetectorRunner "
481                   "(needs videos/ in the corpus; honours detector_impl).")
482     pt.add_argument("--segmenter", default="wasb_hybrid",
483                   help="trajectory hybrid whose detector is used for --trajectory-
source detector")
484     pt.add_argument("--floor", default=None, help="floor baseline JSON (local path or
storage URI)")
485     pt.add_argument("--config", default=None)
486     pt.add_argument("--scratch", default=None)
487     pt.add_argument("--fps", type=float, default=30.0, help="synth trajectory fps
(detector probes the video)")
488     pt.add_argument("--frame-width", type=float, default=1280.0,
489                   help="synth trajectory frame width (detector probes the video)")
490     pt.add_argument("--set", action="append", default=[], metavar="k=v")
491     pt.set_defaults(func=cmd_train)
492     return p
493
494 def _setup_logging(verbose: bool) -> None:
495     """Configure logging for the worker CLI so the pipeline's existing INFO/ERROR
diagnostics are
496     VISIBLE ? the worker previously ran with no logging config, so every

```

```

segmenter/detector log
497         (detector choice, 'Running native WASB inference', availability failures) was
silently dropped.
498         That is exactly why a 0-rally cold-start gave nothing to analyse. Configured here
(the CLI
499         entry), not at import, so importing the module never reconfigures a caller's
logging.""
500         # force=True: basicConfig is a no-op if the root already has handlers (e.g. a
library
501         # configured logging at import), which would leave the level at WARNING and re-
suppress the
502         # diagnostics. Forcing reconfiguration means the worker reliably controls its own
verbosity.
503         logging.basicConfig(
504             level=logging.DEBUG if verbose else logging.INFO,
505             format="%(asctime)s [%(levelname)s] %(name)s: %(message)s",
506             datefmt="%H:%M:%S",
507             force=True,
508         )
509

```

7. user (2026-06-22T10:25:30.350Z)

```

53467c9 docs(data): record stale-label cleanup + set up Phase-2 video copy
6ae3503 docs(data): record Phase-1 GCS migration (trajectories + labels + manifest)
8884c9f docs(data): add DATA_IN_GCS.md ? the corpus source-of-truth map (GCS)
---BRANCH---
work/golden-corpus-hygiene
---PR316 commits on work/golden-corpus-hygiene---
53467c9 docs(data): record stale-label cleanup + set up Phase-2 video copy
6ae3503 docs(data): record Phase-1 GCS migration (trajectories + labels + manifest)
8884c9f docs(data): add DATA_IN_GCS.md ? the corpus source-of-truth map (GCS)
689c738 docs(pkg): mark P2c ? + the loop closed ? v1 LIGHT is feature-complete (#313)
293e5c0 feat(pkg): re-bundle backend/ui ? serve the P2c first-run wizard from indexer --app
(#312)
f8b2c3b feat(pkg): P2b-install ? batteries-included install + truly-local 'motion' first-run
default (gate 08) (#310)
0ed4cc6 fix(segmenter): fail the job on an unrunnable config instead of a silent 0-rally success
(Finding B) (#311)
a42c132 docs(pkg): track the batteries-included app plan (PACKAGING_PLAN.md) + index pointers
(#309)
de2bf14 feat(pkg): P2e ? lazy-torch so the LIGHT tier imports with NO torch installed (default-
OFF) (#308)
679e642 feat(pkg): P2d ? DB upgrade-safety (downgrade guard + backup-before-migrate + export)
(default-OFF) (#307)
f01dc85 feat(db): v7 ? record the UI locale per job (PMF analytics) + count_jobs_by_locale
(#306)
fa7ee3f feat(pkg): P2a ? single ffmpeg/ffprobe resolver (env-overridable; one fail-loud place)
(default-OFF) (#305)
e99d0c3 feat(stitch): localize the reel watermark to the UI locale (+ OFL Noto fonts) (#304)
ca34787 feat(pkg): P2b ? friendly `indexer --app` launcher (boot localhost server + open the
bundled UI) (#303)
4a06b8b feat(pkg): P1b ? bundle the KhelSutra SPA single-origin into the engine wheel (default-
OFF) (#302)
be0c434 feat(pkg): P1a ? net-new console/wizard endpoints: /api/version, /api/capabilities,
/api/videos, /api/reels (#300)
cb7394b feat(serving): T11 ? retention sweep for stale uploads + caches (#299)
c039b90 feat(pkg): P0e ? single-instance lock so a 2nd process can't corrupt in-flight job state
(default-OFF) (#298)
e8bad40 feat(pkg): P0d ? declare scipy (undeclared serving dep) + license-scan & built-wheel CI
lanes (#297)
3895963 feat(pkg): P0c ? config/profile-driven bind address + DNS-rebinding Host-header guard
(default-OFF) (#296)

```

8. assistant (2026-06-22T10:25:34.854Z)

PR #316 is docs-only (3 commits, all `docs(data):`). The branch context note said `work/golden-corpus-hygiene` but the current checkout is on `serving/exposure-safety-preflight` ? those 3 commits sit on top of the merged packaging work. Let me verify the #236 litmus pinning claim and check whether PR #316 is purely docs or also touches scripts.

9. user (2026-06-22T10:25:38.193Z)

.gitattributes	15 +
.github/workflows/ci.yml	84 ++
.gitignore	Bin 924 -> 1117 bytes
CLAUDE.md	19 +-
README.md	32 +
backend/api/__init__.py	1 +
backend/api/auth.py	129 ++++
backend/api/job_worker.py	184 +++++
backend/api/uploads.py	155 ++++
backend/assets/fonts/LICENSE-OFL-1.1.txt	93 +++
.../assets/fonts/NotoSansDevanagari-Variable.ttf	Bin 0 -> 647144 bytes
backend/assets/fonts/NotoSansKannada-Variable.ttf	Bin 0 -> 641372 bytes
backend/assets/fonts/README.md	33 +-
backend/billing.py	60 ++
backend/compute/__init__.py	61 ++
backend/compute/base.py	53 ++
backend/compute/cloud_run.py	155 ++++
backend/config/__init__.py	35 +-
backend/config/loader.py	155 +++-
backend/config/models.py	173 ++++-
backend/config/presets.py	132 ++++
backend/config/startup.py	279 ++++++
backend/eval/ablation.py	162 +--
backend/eval/ablation_pdf.py	141 ++++
backend/eval/experiment.py	82 ++
backend/eval/fusion_audio.py	43 +-
backend/eval/fusion_compare.py	39 +-
backend/eval/golden_fixtures.py	848 ++++++
backend/eval/golden_real_fixtures.py	343 ++++++
backend/eval/rally_seg_eval.py	415 ++++++-
backend/eval/tolerance_metrics.py	277 ++++++
backend/eval/windowing.py	41 +-
backend/infrastructure/database.py	814 ++++++-----
backend/main.py	822 ++++++-----
backend/pipeline/audio/hit_detector.py	4 +-
backend/pipeline/detectors/court_gate.py	99 +++
backend/pipeline/detectors/device.py	56 ++
backend/pipeline/detectors/native_wasb_runner.py	263 ++++++
backend/pipeline/detectors/person_detector.py	18 +-
backend/pipeline/detectors/stub_runner.py	157 ++++
backend/pipeline/detectors/tracknet_runner.py	196 ++++-
backend/pipeline/detectors/wasb_infer.py	43 +-
backend/pipeline/segmenters/_keyhint.py	13 +
backend/pipeline/segmenters/fusion_hybrid.py	29 +-
backend/pipeline/segmenters/gemini.py	568 ++++++-----
backend/pipeline/segmenters/motion.py	30 +-
backend/pipeline/segmenters/trajectory_hybrid.py	68 +-
backend/pipeline/segmenters/wasb_hybrid.py	8 +
backend/pipeline/segmenters/yolo_hybrid.py	161 +--
backend/pipeline/stitcher/compiler.py	230 +----
backend/pipeline/stitcher/watermark_i18n.py	72 ++
backend/pipeline/validators/format.py	3 +-
backend/pipeline/validators/pts_continuity.py	3 +-
backend/providers/gemini.py	4 +
backend/results.py	20 +-
backend/storage/__init__.py	126 +++
backend/storage/base.py	49 ++
backend/storage/gcs.py	63 ++

backend/storage/gdrive.py	135 ++++
backend/storage/local.py	69 ++
backend/storage/ref.py	98 +++
backend/storage/s3.py	78 ++
backend/tools/rally_slicer.py	5 +-
backend/ui/THIRD_PARTY_NOTICES.md	8 +
backend/ui/assets/index-BcE6Y0gS.css	1 +
backend/ui/assets/index-GSXqL6iY.js	60 ++
...evanagari-devanagari-wght-normal-DWI_L5Cu.woff2	Bin 0 -> 121192 bytes
...devanagari-latin-ext-wght-normal-CLKupaxV.woff2	Bin 0 -> 13852 bytes
...ans-devanagari-latin-wght-normal-Bxfr4UJH.woff2	Bin 0 -> 25384 bytes
...sans-kannada-kannada-wght-normal-DuwB-P8d.woff2	Bin 0 -> 90796 bytes
...ns-kannada-latin-ext-wght-normal-MdBBF41x.woff2	Bin 0 -> 13992 bytes
...o-sans-kannada-latin-wght-normal-DeLZCgFT.woff2	Bin 0 -> 27576 bytes
backend/ui/index.html	13 +
backend/ui/ui_version.json	7 +
backend/utls/app_home.py	116 +++
backend/utls/ffmpeg.py	62 ++
backend/utls/paths.py	18 +-
backend/utls/redact.py	51 ++
backend/utls/single_instance.py	76 ++
backend/utls/video.py	6 +-
backend/utls/workspace.py	114 +++
backend/worker/__init__.py	7 +
backend/worker/__main__.py	518 ++++++++
deploy/cloudrun/Dockerfile	61 ++
deploy/cloudrun/README.md	61 ++
deploy/digitalocean/README.md	75 ++
deploy/digitalocean/bootstrap.sh	62 ++
deploy/digitalocean/gpu_setup.sh	69 ++
deploy/digitalocean/mount_scratch.sh	46 ++
deploy/gcp/README.md	257 ++++++
deploy/gcp/create_gpu_vm.sh	58 ++
deploy/gcp/ops-agent-startup.sh	20 +
deploy/gcp/provision.sh	71 ++
deploy/gcp/teardown.sh	47 ++
docs/BACKLOG.md	2 +-
docs/CLOUD_GPU_DRYRUN_GUIDE.md	203 +++++
docs/CODE_MAP.md	15 +-
docs/COMMERCIALIZATION.md	2 +-
docs/COMPUTE_DECOUPLED_SERVING/README.md	170 +++++
docs/COMPUTE_DECOUPLED_SERVING/TESTING_STRATEGY.md	83 ++
docs/COMPUTE_DECOUPLED_SERVING/architecture.svg	130 ++++
docs/COMPUTE_DECOUPLED_SERVING/runlogs/README.md	35 +
.../runlogs/RUNLOG_truly-local_TEMPLATE.md	49 ++
docs/DATA_IN_GCS.md	220 ++++++
docs/DESKTOP_APP_PLAN.md	164 ++++
docs/DOC_STATUS.md	92 +++
docs/EXPERIMENT_HARNESS.md	2 +-
docs/GOLDEN_REGRESSION_FIXTURES.md	216 ++++++
docs/HOSTING_PLAN.md	4 +
docs/HOW_RALLY_DETECTION_WORKS.md	2 +-
docs/I18N_PLAN.md	9 +-
docs/MULTI_SIGNAL_FUSION_PLAN.md	24 +-
docs/NEXT_STEPS.md	59 +-
docs/PACKAGING_PLAN.md	171 +++++
docs/PER_VIDEO_WORKSPACE.md	112 +++
docs/PLATFORM_ARCHITECTURE.md	4 +-
docs/PROJECT_DOC_TEMPLATE.md	74 ++
docs/PROMINENT_COURT_DETECTION.md	126 +++
docs/QUALITY_ITERATIONS.md	378 ++++++-
docs/R1_MILESTONE_LAUNCH_REPORT.md	89 +++
docs/R2_EVAL_METRIC_DESIGN.md	195 +++++
docs/RALLY_DETECTION_GAP_CLOSURE.md	104 +++
docs/RALLY_DETECTION_QUALITY_REPORT.md	478 ++++++++

docs/RALLY_QUALITY_RESEARCH.md	70 +-
docs/README.md	29 +-
docs/REAL_FOOTAGE_VALIDATION_REPORT.md	119 +++
docs/ROADMAP.md	4 +-
docs/SETUP_NEW_MACHINE.md	312 +++++++
docs/UI_PROPOSAL.md	6 +
...R_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md}	12 +-
docs/archives/decisions/DECISIONS.md	205 +++++
.../field-pmf}/FIELD_VISIT_ANSWER_SHEET.md	0
.../field-pmf}/FIELD_VISIT_PLAYBOOK.md	0
docs/{ => archives/field-pmf}/PMF_FIELD_SIGNALS.md	0
.../field-pmf}/PMF_MARKET_RESEARCH.md	0
.../field-pmf}/PRODUCT_FIELD_ASSOCIATE_FLYER.md	0
docs/archives/field-pmf/README.md	15 +
.../DECOUPLED_COMPUTE/01-EXISTING-PLANS-AND-GAP.md	137 ++++
.../02-COMPUTE-MAP-AND-CONTRACT.md	78 ++
.../DECOUPLED_COMPUTE/03-PLATFORM-OPTIONS.md	94 +++
.../DECOUPLED_COMPUTE/04-BUILD-AND-TEST-PLAN.md	161 ++++
.../DECOUPLED_COMPUTE/05-COST-ATTRIBUTION.md	105 +++
.../06-RISKS-AND-OPEN-QUESTIONS.md	85 +++
.../DECOUPLED_COMPUTE/07-COMPUTE-ABSTRACTION.md	168 ++++
.../DEPLOY_REPORT_GCP_2026-06-20.md	110 +++
.../past_projects/DECOUPLED_COMPUTE/HANDOFF.md	108 +++
.../past_projects/DECOUPLED_COMPUTE/README.md	126 +++
docs/archives/past_projects/README.md	3 +-
.../2026-06-18-lovo-resweep-n13/README.md	75 ++
.../2026-06-18-lovo-resweep-n13/lovo_resweep.py	111 +++
docs/archives/quality-iterations/README.md	1 +
.../roora-vendor}/GOLDEN_SET_VENDORING.md	0
docs/archives/roora-vendor/README.md	14 +
.../roora-vendor}/ROORA_LABELING_GUIDELINES.md	0
docs/{ => data_pipeline}/GOLDEN_DATA_SHARING.md	2 +-
.../GOLDEN_SET_IMPLEMENTATION.md	2 +-
.../GOLDEN_SET_PHASE1_VIDEOS.md	2 +-
docs/data_pipeline/GOLDEN_VIDEOS.md	60 ++
docs/data_pipeline/README.md	16 +
.../VIDEO_RECORDING_GUIDELINES.md	21 +
eval_baselines/fixtures/README.md	70 ++
.../fixtures/clean_single_rally/expected.json	39 +
.../fixtures/clean_single_rally/labels.csv	2 +
.../fixtures/clean_single_rally/provenance.json	23 +
.../fixtures/clean_single_rally/trajectory.csv	121 +++
eval_baselines/fixtures/fixtures/manifest.json	62 ++
.../multi_rally_with_deadtime/expected.json	54 ++
.../fixtures/multi_rally_with_deadtime/labels.csv	3 +
.../multi_rally_with_deadtime/provenance.json	23 +
.../multi_rally_with_deadtime/trajectory.csv	331 +++++++
.../fixtures/occlusion_break/expected.json	54 ++
eval_baselines/fixtures/occlusion_break/labels.csv	3 +
.../fixtures/occlusion_break/provenance.json	23 +
.../fixtures/occlusion_break/trajectory.csv	331 +++++++
eval_baselines/nightly_baseline.json	569 ++++++
mypy.ini	13 +
pyproject.toml	17 +
requirements.txt	1 +
scripts/bundle_ui.sh	95 +++
scripts/doctor.py	31 +-
scripts/install.py	225 +++++
scripts/nightly_regression.ps1	83 ++
scripts/nightly_regression.py	633 ++++++
scripts/register_nightly_task.ps1	70 ++
scripts/run_phase3_eval.py	11 +-
scripts/run_process_and_stitch.py	23 +-
scripts/uninstall.py	179 +++++
tests/test_ablation.py	47 +-

tests/test_api_endpoints.py	406 ++++++++
tests/test_app_home.py	187 +++++
tests/test_app_launcher.py	170 +++++
tests/test_async_jobs.py	8 +-
tests/test_audio_features.py	26 +
tests/test_billing.py	192 +++++
tests/test_bind_and_host.py	75 ++
tests/test_cleanup_caches.py	78 ++
tests/test_compute_backend.py	289 ++++++
tests/test_config.py	28 +-
tests/test_config_deployment.py	371 ++++++++
tests/test_config_wasb.py	16 +-
tests/test_court_gate.py	103 +++
tests/test_database.py	170 ++++
tests/test_db_upgrade_safety.py	78 ++
tests/test_device_context.py	47 ++
tests/test_doctor.py	67 ++
tests/test_edge_auth.py	231 ++++++
tests/test_experiment.py	72 ++
tests/test_ffmpeg_resolver.py	78 ++
tests/test_fusion_compare.py	25 +
tests/test_fusion_hybrid.py	10 +
tests/test_gemini.py	37 +
tests/test_generate_highlights.py	174 ++++
tests/test_golden_fixtures.py	138 ++++
tests/test_golden_fixtures_split.py	87 +++
tests/test_golden_real_fixtures.py	504 ++++++++
tests/test_input_source.py	152 ++++
tests/test_job_worker_loop.py	8 +-
tests/test_job_worker_module_seam.py	97 +++
tests/test_light_tier_no_torch.py	52 ++
tests/test_locale_analytics.py	84 ++
tests/test_motion_segmenter.py	222 ++++++
tests/test_native_wasb_runner.py	383 ++++++++
tests/test_nightly_regression.py	343 ++++++++
tests/test_output_base.py	90 +++
tests/test_p2b_install.py	243 ++++++
tests/test_person_detector.py	14 +-
tests/test_profile_parity.py	140 ++++
tests/test_provider_gemini.py	21 +
tests/test_rally_seg_eval.py	155 +++-
tests/test_rally_seg_eval_report.py	306 ++++++++
tests/test_redact.py	78 ++
tests/test_results.py	50 ++
tests/test_served_gate_a_regression.py	96 +-
tests/test_server_entrypoint.py	177 +++++
tests/test_single_instance.py	54 ++
tests/test_startup_checks.py	369 ++++++++
tests/test_storage.py	328 ++++++
tests/test_stub_runner.py	129 ++++
tests/test_tolerance_metrics.py	188 +++++
tests/test_tracknet_runner.py	96 ++
tests/test_training_gen0.py	56 ++
tests/test_ui_bundle.py	54 ++
tests/test_upload_download.py	3 +-
tests/test_wasb_cache.py	20 +
tests/test_watermark_i18n.py	94 +++
tests/test_windowing.py	37 +-
tests/test_worker.py	309 ++++++++
tests/test_workspace.py	122 +++
tests/test_yolo_hybrid.py	41 +
tools/cleanup_caches.py	173 ++++
tools/generate_highlights.py	118 +-
training/gen0/harness.py	50 +-

252 files changed, 26179 insertions(+), 1466 deletions(-)

```

---scratch script tracked?---
(empty = gitignored, not tracked)
---#236 litmus path refs---
backend/config/models.py
backend/eval/ablation.py
backend/eval/calibrate_local.py
backend/eval/distill_local.py
backend/eval/eval_partial_labels.py
backend/eval/experiment.py
backend/eval/export_wasb_segments.py
backend/eval/fusion_golden.py
backend/eval/rally_seg_eval.py
backend/eval/served_gate_a.py

```

10. assistant (2026-06-22T10:25:43.483Z)

The `master..work/golden-corpus-hygiene` diff is huge because the local `master` lags origin (the packaging PRs aren't in local master). The PR #316-specific commits are the 3 `docs(data):` ones. Let me confirm the served Gate-A litmus path-pinning and the golden-fixtures manifest path handling ? the two cross-thread coupling points.

11. user (2026-06-22T10:25:44.982Z)

```

1      """SERVED Gate-A verification: run the *production* ``FusionSegmenter`` decision seems
2      on a cached WASB trajectory and score the resulting AI-handoff windows vs golden labels.
3
4      The offline litmus (``backend.eval.ablation`` / ``fusion_compare``) measures Gate A on
the
5      **permissive proposal windowing** (``distill_local.PROPOSAL`` = recall-oriented) re-
scored
6      through the experiment harness's ``_gate_mask``. That validates the *signal*. It does
NOT
7      exercise the **served path**: in production the windows come from
8      ``trajectory_to_action_windows`` at the production operating point
9      (``velocity_threshold=0.02, dead_time_merge_gap=2.0, min_action_window_duration=2.0``),
the
10     net-crossing count is computed by ``FusionSegmenter._enrich_candidate_stats``, and the
gate is
11     ``FusionSegmenter._select_for_handoff``. THIS module drives exactly those production
methods ?
12     no reimplementations of the windowing or the gate ? on a cached trajectory CSV, so the
13     verification reflects what ``process_video`` actually hands to Gemini.
14
15     GPU-free and Gemini-free: it stops at the handoff-selection boundary (the windows that
WOULD
16     be sent to the AI). With the person cue OFF (the default) nothing here loads torch or
touches
17     the GPU ? it just re-scores cached WASB trajectories.
18
19     Run (owner box, golden trajectories present):
20     python -m backend.eval.served_gate_a --manifest output/human_lovo_manifest.json
21     """
22     from __future__ import annotations
23
24     import argparse
25     import json
26     from dataclasses import dataclass
27     from typing import Any, Dict, List, Optional, Tuple
28
29     from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
30     from backend.pipeline.segmenters.fusion_hybrid import FusionSegmenter
31     from backend.eval.calibrate_wasb import load_golden_gts
32     from backend.eval.metrics import Interval, score, Score
33     from backend.eval import segmentation_metrics as _segm
34

```

```

35     # Production windowing operating point (config.json indexing defaults ? the values
36     # TrajectoryHybridSegmenter.process_video reads for the served fusion run).
37     PROD_VELOCITY_THRESH = 0.02
38     PROD_MERGE_GAP = 2.0
39     PROD_MIN_WINDOW = 2.0
40
41
42     def _served_idx_cfg(min_crossings: int) -> Dict[str, Any]:
43         """The ``indexing`` config dict the served FusionSegmenter seams read ? mirrors
44         config.json's ``indexing.fusion`` block, with Gate A at ``min_crossings`` and the
45         person cue OFF (so no GPU / torch is touched)."""
46         return {
47             "fusion": {
48                 "substrate": "wasb",
49                 "min_crossings": min_crossings,
50                 "min_players": 0,
51                 "cue_flags": {},
52             },
53             "wasb": {"velocity_threshold": PROD_VELOCITY_THRESH},
54             "dead_time_merge_gap": PROD_MERGE_GAP,
55             "min_action_window_duration": PROD_MIN_WINDOW,

```

12. user (2026-06-22T10:25:46.788Z)

(Bash completed with no output)

13. assistant (2026-06-22T10:25:51.554Z)

The grep returned nothing ? neither `served\_gate\_a.py`, `golden\_real\_fixtures.py`, nor `run\_phase3\_eval.py` contain any `gs://`/`gcs://`/`input\_root`/`resolve\_input\_uri` handling. That confirms the litmus and fixtures are local-path only; flipping them to GCS is genuine net-new wiring, not a config flip. Let me confirm how the litmus default manifest path resolves and check the phase3 eval script.

14. user (2026-06-22T10:25:52.084Z)

```

1     """Phase 3 driver: batch process + evaluate a collector export against ground truth.
2
3     Walks the `manifest.json` produced by sports-data-collector's `curate export`,
4     processes each video through a segmenter (so detections land in the indexer DB),
5     scores each against its ground-truth rally CSV, and prints per-video plus
6     corpus-aggregate precision/recall/F1, mean IoU, and boundary error.
7
8     Examples
9     -----
10    Free CPU baseline on a single video (validate the loop):
11        python run_phase3_eval.py --manifest ../sports-data-
12        collector/data/eval_export/manifest.json \
13        --segmenter motion --only shuttlest_1
14
15    Full corpus with the two-stage detector (needs GEMINI_API_KEY; costs API calls):
16        python run_phase3_eval.py --manifest ../manifest.json --segmenter yolo_hybrid \
17        --json output/phase3_report.json
18
19    Re-score only (videos already processed), no re-segmentation:
20        python run_phase3_eval.py --manifest ../manifest.json --score-only
21    """
22
23    import os
24    import sys
25    import json
26    import time
27    import argparse
28    import logging

```



```

28
29     from dotenv import load_dotenv
30     load_dotenv() # make GEMINI_API_KEY (etc.) available to the AI-backed segmenters
31
32     from backend.infrastructure.database import Database
33     from backend.config import load_config
34     from backend.utils.hashing import compute_video_id as get_file_md5 # canonical video_id
35     # Importing the package registers all segmenters (motion/gemini/yolo_hybrid/...).
36     from backend.pipeline.segmenters import segmenter_registry
37     from backend.results import unwrap_or_failure
38     from backend.eval.annotations import load_annotations
39     from backend.eval.harness import evaluate, format_report_text, report_to_dict
40     from backend.eval import batch
41
42     logger = logging.getLogger(__name__)
43
44
45     def build_parser() -> argparse.ArgumentParser:
46         p = argparse.ArgumentParser(
47             description="Batch process + evaluate a collector export manifest (Phase 3).",
48             formatter_class=argparse.RawDescriptionHelpFormatter,
49         )
50         p.add_argument("--manifest", required=True, help="Path to the collector's
manifest.json.")
51         p.add_argument("--videos-root", default=None,
52             help="Root for resolving manifest relative local_paths "
53             "(default: the collector repo root, two levels above the
manifest).")
54         p.add_argument("--segmenter", default="motion",
55             help="Segmenter to run (default: motion ? free/CPU). "
56             "yolo_hybrid/gemini need GEMINI_API_KEY and cost API calls.")
57         p.add_argument("--stage", choices=("candidates", "final", "both", "auto"),
58             default="auto",
59             help="Which prediction stage(s) to score (default: auto by
segmenter).")
60         p.add_argument("--detector", default=None, help="Restrict candidate scoring to this
detector.")
61         p.add_argument("--iou", type=float, default=0.5, help="IoU match threshold (default:
0.5).")
62         p.add_argument("--only", action="append", default=None,
63             help="Only this video_id (repeatable).")
64         p.add_argument("--limit", type=int, default=None, help="Process at most N videos.")
65         p.add_argument("--max-frames", type=int, default=None,
66             help="Cap segmenter frames (fast validation).")
67         p.add_argument("--score-only", action="store_true",
68             help="Skip segmentation; only score detections already in the DB.")
69         p.add_argument("--reprocess", action="store_true",
70             help="Re-run segmentation even if the video already has segments.")
71         p.add_argument("--db", default=None, help="Indexer DB path (default:
output/<db_name>).")
72         p.add_argument("--json", dest="json_out", default=None, help="Write a JSON report
here.")
73         p.add_argument("--show-failures", action="store_true", help="List missed/spurious
rallies per video.")
74         return p
75
76     def main(argv=None) -> int:
77         logging.basicConfig(level=logging.INFO, format="%(levelname)s %(message)s")
78         args = build_parser().parse_args(argv)
79
80         manifest_path = os.path.abspath(args.manifest)
81         if not os.path.exists(manifest_path):
82             print(f"ERROR: manifest not found: {manifest_path}", file=sys.stderr)
83             return 2

```

```

84     manifest_dir = os.path.dirname(manifest_path)
85     videos_root = args.videos_root or os.path.dirname(os.path.dirname(manifest_dir))
86
87     with open(manifest_path) as f:
88         manifest = json.load(f)
89     entries = manifest.get("eval_sets", [])
90
91     cfg = load_config() # typed AppConfig (same object the live pipeline feeds
segmenters)
92     db_path = args.db or os.path.join(cfg.output.output_dir, cfg.output.db_name)
93     os.makedirs(os.path.dirname(db_path) or ".", exist_ok=True)
94     db = Database(db_path)
95
96     stages = batch.default_stages_for(args.segmenter, args.stage)
97
98     # Resolve the segmenter class up front (unless purely scoring).
99     segmenter = None
100    if not args.score_only:
101        seg_cls = segmenter_registry.get(args.segmenter)
102        if not seg_cls:
103            print(f"ERROR: segmenter '{args.segmenter}' not found. "
104                  f"Available: {list(segmenter_registry.list_available().keys())}",
file=sys.stderr)
105            return 2
106        segmenter = seg_cls(db, cfg)
107
108    # Filter entries.
109    targets = []
110    for e in entries:
111        if args.only and e["video_id"] not in args.only:
112            continue
113        targets.append(e)
114    if args.limit:
115        targets = targets[: args.limit]
116
117    per_video = []
118    skipped = []
119    print(f"Phase 3: {len(targets)} target(s) | segmenter={args.segmenter} |
stages={stages} | db={db_path}\n")
120
121    for e in targets:
122        vid = e["video_id"]
123        video_path = batch.resolve_video_path(e.get("local_path"), videos_root)
124        csv_path = os.path.join(manifest_dir, e["csv"])
125
126        if not video_path or not os.path.exists(video_path):
127            skipped.append((vid, "no video file"))
128            print(f"[skip] {vid}: no video file ({e.get('local_path')})")
129            continue
130        if not os.path.exists(csv_path):
131            skipped.append((vid, "no GT csv"))
132            print(f"[skip] {vid}: GT csv missing ({csv_path})")
133            continue
134
135        video_id = get_file_md5(video_path)
136
137        # Segment unless score-only or already present (and not --reprocess).
138        if not args.score_only:
139            already = db.query_play_segments(video_id, {}) or
db.query_candidate_segments(video_id)
140            if already and not args.reprocess:
141                print(f"[have] {vid}: detections already in DB (use --reprocess to
redo)")
142            else:
143                print(f"[proc] {vid}: segmenting with {args.segmenter}...")

```

```

144         db.add_video(video_id, video_path, "processed", [])
145         t0 = time.time()
146         # process_video returns a Result (Success|Failure) for some segmenters
147         # (e.g. the default `motion`); unwrap before len() so this doesn't crash
148         with
149             # TypeError on a Success dataclass ? mirrors the unwrap in
150             backend/main.py.
151             segs, failure = unwrap_or_failure(
152                 segmenter.process_video(video_id, video_path,
153                 max_frames=args.max_frames))
154             if failure is not None:
155                 print(f"          -> FAILED: {failure.message}")
156                 skipped.append((vid, f"segment failed: {failure.message}"))
157                 continue
158                 print(f"          -> {len(segs)} segments in {time.time() - t0:.0f}s")
159
160         gt = load_annotations(csv_path)
161         report = evaluate(db, video_id, gt, stages, iou_threshold=args.iou,
162         detector=args.detector)
163         print(format_report_text(report, show_failures=args.show_failures))
164         print()
165         d = report_to_dict(report)
166         d["collector_video_id"] = vid
167         per_video.append(d)
168
169         agg = batch.aggregate(per_video, stages)
170         print(batch.format_aggregate(agg))
171         if skipped:
172             print(f"\nSkipped {len(skipped)}: " + ", ".join(f"{v} ({why})" for v, why in
173             skipped))
174
175         if args.json_out:
176             with open(args.json_out, "w") as f:
177                 json.dump({"segmenter": args.segmenter, "iou": args.iou, "stages": stages,
178                 "aggregate": agg, "per_video": per_video,
179                 "skipped": skipped}, f, indent=2)
180             print(f"\nJSON report written to {args.json_out}")
181
182         return 0
183
184     if __name__ == "__main__":
185         raise SystemExit(main())
186
187

```

15. assistant (2026-06-22T10:25:57.638Z)

This confirms the key foresight point: `run\_phase3\_eval.py` uses `os.path.abspath`, `os.path.exists`, `os.path.join`, and `batch.resolve\_video\_path` against local filesystem paths throughout ? it does NOT route through `resolve\_input\_uri` or any storage backend. Same for `served\_gate\_a.py` (reads a local manifest + `load\_golden\_gts` on local CSVs). Phase-3 "flip to GCS" is real plumbing work, not an env override.

I have enough grounded evidence. Let me compile the prioritized foresight verdict.

16. user (2026-06-22T10:26:42.714Z)

Structured output provided successfully

17. assistant (2026-06-22T10:26:45.340Z)

Structured verdict delivered.